(16) DESIGN WORD PROCESSOR :- FLYWEIGHT PATTERN.

→ structural Pattern helps to reduce the memory usage by sharing data across multiple objects.

→ When it is used ?

* memory is Limited.
* Object shared same data.
  → Intrinsic = same data
  → Extrinsic = different data
* Creation of object is expensive.

GAME example

① Create 1000 human robot →] Total = 2000
② Create 1000 Dog robot

```
class Robot {
  x axis
  y axis
  type
  Sprite imageBody
  Robot() {
  }
}

main() { x=0, y=0.
  for (i = 1 -> 1000) {
    Sprite human = new Sprite()
    Robot human = new Robot(x+i, y+i, "human", "human")
  }
  for (i = 1 -> 1000) {
    Sprite Dog = new Sprite()
    Robot Dog = new Robot(____);
```

⇒ If each object is taking 1KB of memory total is 2000KB

OBSERVATIONS

⇒ There is Common data in all object = type & image

⇒ Uncommon data = X axis & Y axis.

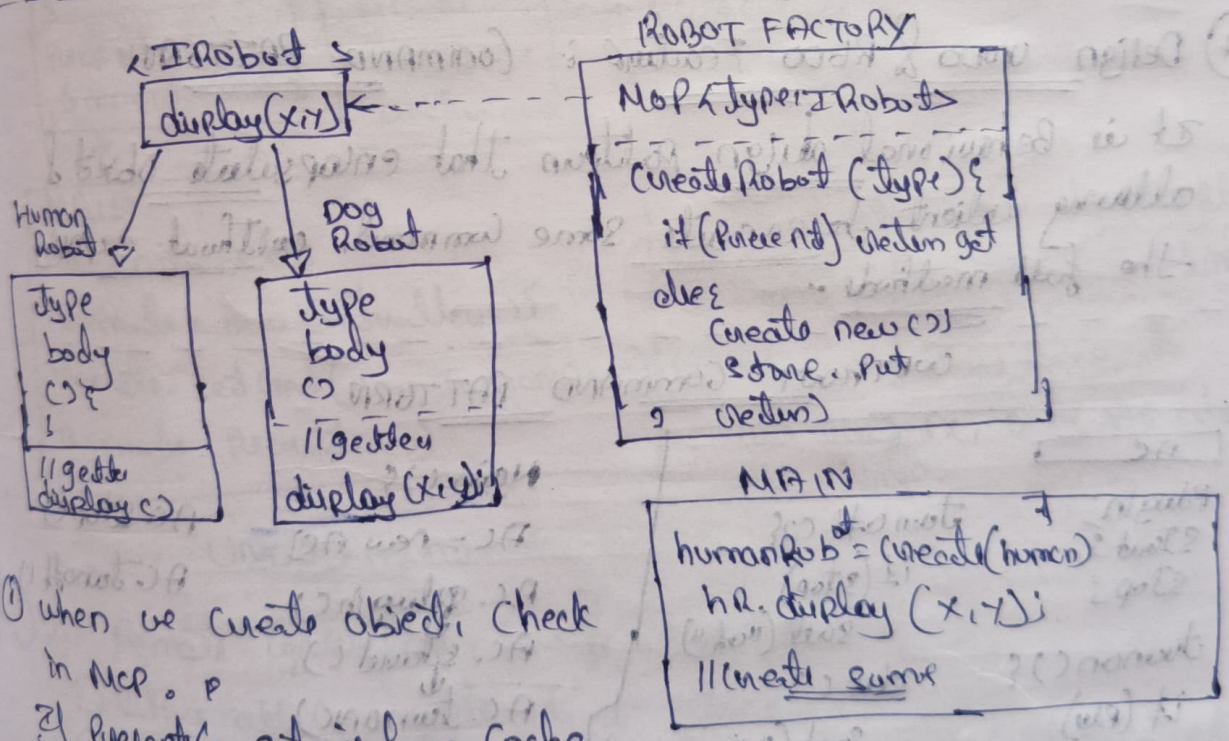
How to use FLYWEIGHT & Relate IT ?

(1) From OBJECT, remove Ext Extrinsic data & keep Intrinsic Data. (This is FLYWEIGHT Object)

(2) Make FLYWEIGHT class as Immutable.

(3) Pass extrinsic data through method () Parameters

(4) Once Flyweight Object is Created, Cache it & reuse.



(1) When we create object, Check in Map.

(2) Present, get it from Cache else create new & store it

(3) Pass X & Y & display the Robots.

(4) Run 2nd time, it won't create new Human Robot

WORD PROCESSOR:-

⇒ Display Sentence = "Hi, How are you".

+ All characters as same, font, size...

① Instead of creating object for all characters, Cache it & reuse the same characters.

